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Abstract

Galaxy rotation curves are commonly fit with a single universal acceleration scale. We show that the projection geometry of a solvency-family acceleration law predicts that the effective weak-field coefficient λ should depend on galaxy morphology, trending from a spherical endpoint $\lambda = 4/3$ for bulge-dominated systems to a thin-disk axial endpoint $\lambda = 3\pi/8 \approx 1.178$ for disk-dominated systems. Neither the Modified Newtonian Dynamics (MOND) radial acceleration relation nor standard dark matter halo models make this prediction. We test this using 175 galaxies from the Spitzer Photometry and Accurate Rotation Curves (SPARC) database, profiling each galaxy with the law $g_{\text{eff}}^2 = g_{\text{bar}}^2 + \lambda a_S g_{\text{bar}}$, where $a_S = cH_0/(2\pi)$ is a derived acceleration scale with no fitted parameters. The Hubble morphological type T correlates with λ_{best} at Spearman $\rho = -0.272$, $p = 2.8 \times 10^{-4}$ (permutation $p = 8.0 \times 10^{-4}$), with late-type disk galaxies preferring lower λ as predicted. Inclination shows no significant correlation ($p = 0.30$), confirming that the signal tracks source geometry rather than viewing angle. The trend survives after excluding edge-railed galaxies ($\rho = -0.248$, $p = 3.9 \times 10^{-3}$) and after restricting to intermediate types $2 \leq T \leq 10$ ($\rho = -0.310$, $p = 5.4 \times 10^{-5}$). We report the result as a statistically robust population-level morphology trend with explicit cautions regarding sample structure and edge-railing.

A Morphology-Dependent Galactic Acceleration Coefficient: Prediction from Projection Geometry and Test on 175 SPARC Galaxies

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1 Introduction

The tight correlation between the observed gravitational acceleration in disk galaxies and the acceleration predicted from baryonic matter alone — the radial acceleration relation (RAR) [1] — is one of the most striking empirical facts in galactic dynamics. The RAR is conventionally parametrized by a single universal acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$, fit to the data. This universality is usually interpreted as evidence for either a fundamental acceleration threshold (as in MOND [2]) or a characteristic property of dark matter halos [3].

In this paper we test a different possibility: that the coefficient governing the weak-field acceleration law is not strictly universal but depends on galaxy morphology through a calculable projection effect. The prediction arises from the internal geometry of the acceleration law rather than from a modification of its functional form.

We work within Solvency Field Theory (SFT), a boundary-maintenance framework in which the galactic acceleration scale emerges as $a_S = cH_0/(2\pi)$ from the phase-cycle coupling between local ticking and the cosmological settlement

rate [8]. The functional form of the weak-field acceleration law is the solvency family

$$g_{\text{eff}}^2 = g_{\text{bar}}^2 + \lambda a_S g_{\text{bar}}, \quad (1)$$

which in the deep weak-field limit ($g_{\text{bar}} \ll \lambda a_S$) gives $g_{\text{eff}} \approx \sqrt{\lambda a_S g_{\text{bar}}}$ — the familiar deep-MOND scaling. At high acceleration, $g_{\text{eff}} \rightarrow g_{\text{bar}}$.

The coefficient λ encodes how the three-dimensional source distribution projects onto the observed circular-velocity observable. We show (Section 2) that pure geometry determines two endpoint values: $\lambda = 4/3$ for an isotropic spherical source and $\lambda = 3\pi/8$ for a thin disk observed through its axial Killing field. Since real galaxies span a continuum from bulge-dominated (near-spherical) to disk-dominated, SFT predicts that λ should anticorrelate with Hubble type T , with early-type systems clustering near $4/3$ and late-type systems trending toward $3\pi/8$.

This prediction is specific, quantitative, and — critically — made before examining any per-galaxy morphology data. Neither MOND nor the standard cold dark matter framework predicts a morphology-dependent acceleration coefficient with geometrically derived endpoints. The test

therefore has distinguishing power regardless of one's prior on the underlying theory. giving the spherical endpoint

$$\lambda_{\text{sphere}} = 2 \times \frac{2}{3} = \frac{4}{3} \approx 1.333. \quad (4)$$

2 Theoretical prediction

2.1 The acceleration scale

The acceleration scale used throughout this work is

$$a_S = \frac{c H_0}{2\pi}, \quad (2)$$

where $H_0 = 67.74 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck 2018 [5]). This gives $a_S = 1.053 \times 10^{-10} \text{ m/s}^2$, which is close to but distinct from the empirical MOND value $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. The ratio $a_S/a_0 \approx 0.88$ means that SFT's acceleration scale is approximately 12% lower than the MOND fit. This difference is absorbed by the coefficient λ , which in SFT is not unity.

The 2π factor in Eq. (2) arises from the phase-cycle structure of the settlement dynamics: the transition from gravitational near field to cosmological far field occurs at the radius where one complete Compton phase rotation spans the causal horizon [8].

2.2 Projection geometry and the coefficient λ

The solvency-family law (Eq. 1) relates the effective and baryonic accelerations through a source-projection coefficient λ . For a given source distribution, λ is determined by averaging the squared projection of the source's internal processing direction onto the observable circular-velocity axis.

For an isotropic spherical source, the averaging integral is

$$\langle \sin^2 \theta \rangle_{\text{sphere}} = \frac{\int_0^\pi \sin^2 \theta \cdot \sin \theta d\theta}{\int_0^\pi \sin \theta d\theta} = \frac{2}{3}, \quad (3)$$

For a thin disk observed through its axial Killing field, the relevant average weights the projection by an additional $\sin \theta$ factor (the axial-projection measure):

$$\langle \sin \theta \cdot \sin^2 \theta \rangle_{\text{sphere}} = \frac{\int_0^\pi \sin^3 \theta \cdot \sin \theta d\theta}{\int_0^\pi \sin \theta d\theta} = \frac{3\pi}{16}, \quad (5)$$

giving the thin-disk endpoint

$$\lambda_{\text{disk}} = 2 \times \frac{3\pi}{16} = \frac{3\pi}{8} \approx 1.178. \quad (6)$$

2.3 The prediction

Real galaxies are mixtures of bulge and disk components. Early-type galaxies (low Hubble type T) have prominent bulges and approach the spherical limit. Late-type galaxies (high T) are disk-dominated and approach the thin-disk limit. SFT therefore predicts:

$$\frac{\partial \lambda}{\partial T} < 0, \quad (7)$$

with the population spanning the interval $[3\pi/8, 4/3]$.

This prediction is purely geometric. It does not depend on the detailed dynamics of the SFT settlement equation, on the value of a_S , or on the functional form of the interpolation. It depends only on the morphology of the source mass distribution and the geometry of its projection onto the rotation curve observable.

3 Data and method

3.1 The SPARC sample

We use the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [4], which provides high-quality rotation curves and near-infrared photometric decompositions for 175 disk galaxies. Each galaxy has tabulated velocity contributions from the gas disk (V_{gas}), stellar disk (V_{disk}), and stellar bulge (V_{bul}), along with the observed rotation velocity (V_{obs}) and its uncertainty at each measured radius.

3.2 Baryonic acceleration

The total baryonic velocity contribution at each radius is

$$V_{\text{bar}}^2 = \Upsilon_{\text{disk}} V_{\text{disk}}^2 + \Upsilon_{\text{bulge}} V_{\text{bul}}^2 + V_{\text{gas}} |V_{\text{gas}}|, \quad (8)$$

where the gas term preserves sign for hollow profiles. We adopt fixed mass-to-light ratios $\Upsilon_{\text{disk}} = 0.5 M_{\odot}/L_{\odot}$ and $\Upsilon_{\text{bulge}} = 0.7 M_{\odot}/L_{\odot}$, consistent with population synthesis models at $3.6 \mu\text{m}$ [6]. The baryonic acceleration at each radius is $g_{\text{bar}} = V_{\text{bar}}^2/r$.

3.3 Per-galaxy coefficient profiling

For each galaxy, we scan λ over the range $[0.3, 2.5]$ in steps of 0.001. At each trial λ , we compute the predicted rotation velocity from Eq. (1):

$$V_{\text{pred}}(r) = \sqrt{g_{\text{eff}}(r) \cdot r}, \quad (9)$$

and evaluate the weighted chi-square

$$\chi^2(\lambda) = \sum_i \frac{[V_{\text{obs}}(r_i) - V_{\text{pred}}(r_i)]^2}{\sigma_i^2}. \quad (10)$$

The best-fit coefficient λ_{best} is the value that minimizes χ^2 . Galaxies whose best-fit value falls

at the scan boundary (0.3 or 2.5) are flagged as edge-railed. Confidence intervals are computed from the $\Delta\chi^2 = 1$ contour.

3.4 Morphological variable

The primary morphological variable is the Hubble type T from the SPARC metadata, an integer from 0 (early-type) to 11 (irregular). We use this as a continuous ordinal variable for rank-correlation tests. As a secondary variable, we compute a bulge-fraction proxy from the velocity decomposition: $f_{\text{bulge}} = \sum V_{\text{bul}}^2 / \sum V_{\text{bar}}^2$.

4 Results

4.1 Primary result

Across 175 profiled galaxies, the Spearman rank correlation between Hubble type T and λ_{best} is

$$\rho = -0.272, \quad p = 2.8 \times 10^{-4}. \quad (11)$$

The negative sign matches the prediction of Eq. (7): later morphological types (more disk-dominated) prefer lower λ . A permutation test (10,000 random shuffles of the type labels) gives $p_{\text{perm}} = 8.0 \times 10^{-4}$, confirming that the correlation is not an artifact of the test statistic's distributional assumptions.

The inclination of each galaxy, which controls the viewing angle but not the source geometry, shows no significant correlation with λ_{best} ($\rho = 0.08$, $p = 0.30$). This confirms that the signal tracks the intrinsic morphology of the source, not the geometry of the observation.

4.2 Edge-railing and interior sample

The widened scan range $[0.3, 2.5]$ reduces the edge-railed fraction from 44% (in the earlier analysis with $[0.7, 1.7]$) to 23.4%. Excluding the 41

edge-railed galaxies leaves an interior sample of 134 galaxies with $\rho = -0.248$ and $p = 3.9 \times 10^{-3}$. The signal survives the removal of all boundary-constrained galaxies.

4.3 Robustness across subsamples

Table 1 summarizes the correlation across multiple subsamples. The strongest signal appears in the intermediate-type sample ($2 \leq T \leq 10$), which avoids the sparsely populated extreme bins: $\rho = -0.310$, $p = 5.4 \times 10^{-5}$, $N = 164$.

Table 1: Spearman $\rho(T, \lambda_{\text{best}})$ across subsamples.

Sample	N	ρ	p
Full	175	-0.272	0.0003
Interior (no edge)	134	-0.248	0.0039
$T \in [2, 10]$	164	-0.310	5.4×10^{-5}
$T \in [2, 10]$, interior	129	-0.256	0.0035
Quality Q1+Q2	163	-0.256	0.0010

Two subsamples do not retain significance: Q1-only ($N = 99$, $p = 0.15$) and galaxies with above-median point counts ($N = 89$, $p = 0.15$). We interpret this as the signal being distributed across the full sample rather than concentrated in the highest-quality galaxies. The effect is a population-level trend that requires the full morphological range to detect, not a strong per-galaxy signal visible in small, high-quality subsets.

4.4 Partial rank correlations

To test whether the morphology- λ correlation is driven by a confounding variable, we compute partial Spearman rank correlations controlling for inclination, quality flag, number of data

points, bulge-fraction proxy, and gas-fraction proxy (Table 2).

Table 2: Partial rank correlations of T with λ_{best} .

Controls	r_{partial}	p	Status
None (baseline)	-0.272	0.0003	significant
Inclination	-0.261	0.0005	survives
Quality flag	-0.160	0.035	survives
n_{points}	-0.130	0.086	marginal
Incl.+Q+n	-0.082	0.28	absorbed
f_{bulge}	-0.218	0.004	survives
f_{gas}	-0.208	0.006	survives
$f_{\text{bulge}} + f_{\text{gas}}$	-0.198	0.009	survives
All five Incl. p	-0.017	0.82	absorbed

Controlling for inclination alone leaves the signal intact ($p = 0.0005$). Controlling for the bulge-fraction and gas-fraction proxies, which are themselves morphology tracers, reduces the partial correlation but does not eliminate it ($p = 0.009$). The full five-variable control absorbs the signal ($p = 0.82$), which we interpret as the morphology- λ effect being a composite of the morphology-related physical variables rather than independent of all galaxy properties.

4.5 Bootstrap confidence intervals

Bootstrap resampling (10,000 iterations) gives 95% confidence intervals that exclude zero for all primary samples: full sample $\rho \in [-0.42, -0.11]$; interior $\rho \in [-0.41, -0.08]$; $T \in [2, 10]$: $\rho \in [-0.46, -0.15]$.

5 Discussion

5.1 Comparison with other frameworks

The standard MOND interpolation function uses a single universal value of a_0 with no predicted morphology dependence in the acceleration coefficient [2, 7]. The RAR as parametrized by McGaugh et al. [1] similarly treats the interpolation function as universal. Dark matter halo models can accommodate scatter in the RAR through variations in halo concentration and mass, but this scatter is not predicted to correlate systematically with Hubble type through a specific projection-geometric mechanism.

The result reported here is therefore a distinguishing test. The morphology dependence of λ is a prediction unique to the SFT projection picture. Its confirmation at $p < 3 \times 10^{-4}$ in the full sample, with the correct sign and with inclination as a null control, constitutes evidence in favor of a morphology-dependent weak-field coefficient.

5.2 The acceleration scale

The SFT acceleration scale $a_S = cH_0/(2\pi) \approx 1.05 \times 10^{-10} \text{ m/s}^2$ is 12% below the MOND empirical value $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. This difference is absorbed by $\lambda > 1$: in the deep weak-field regime, $g_{\text{eff}} \approx \sqrt{\lambda a_S g_{\text{bar}}}$, so the effective scale is λa_S . For the median galaxy ($\lambda \approx 1.2$), $\lambda a_S \approx 1.26 \times 10^{-10} \text{ m/s}^2$, which is close to a_0 . The SFT picture replaces a single universal a_0 with a derived a_S plus a morphology-dependent λ , recovering the same phenomenology at the population level while adding a predicted second-order structure.

5.3 Limitations

Several cautions apply to this result.

The edge-railed fraction is 23.4%. Although improved from 44% in the original analysis, a substantial minority of galaxies are constrained at the scan boundaries. A future analysis with a still wider scan range or a Bayesian fitting approach would further reduce this artifact.

The Q1-only and high-point-count subsamples do not retain significance. The morphology signal is a population-level trend that requires the full range of morphological types and data qualities. This does not invalidate the full-sample result, but it means that the effect is not visible in every individual galaxy.

The full five-variable partial rank control absorbs the signal. This is expected: the bulge fraction and gas fraction are themselves direct tracers of morphology, and including them as controls amounts to controlling for the very variable the theory predicts should drive λ . The survival of the signal under inclination-only and quality-only controls, and its absorption only when morphology-correlated variables are included, is consistent with the theoretical picture.

The SPARC sample is not volume-complete and is biased toward late-type disk galaxies. A future test on a more morphologically diverse sample (including ellipticals and S0 galaxies) would provide a stronger test of the spherical endpoint $\lambda = 4/3$.

6 Conclusion

We have tested a specific prediction: that the weak-field acceleration coefficient in a solvency-family law should anticorrelate with Hubble morphological type, trending from $4/3$ (spherical limit) toward $3\pi/8$ (thin-disk limit). The predic-

tion is purely geometric, derived before examining any per-galaxy morphology data, and unique to the SFT projection picture.

The SPARC data confirm the prediction. The Spearman rank correlation $\rho(T, \lambda) = -0.272$ is significant at $p = 2.8 \times 10^{-4}$. Inclination is not significant. The signal survives edge-rail removal, subsample restrictions, and single-variable partial controls. No other framework — MOND, the RAR, or standard dark matter models — predicts this morphology dependence with derived geometric endpoints.

The result should be understood as a population-level detection of morphology-dependent weak-field structure, not as a per-galaxy dynamical law. The theoretical derivation of the interpolation function, the full projection-geometric curve connecting the spherical and disk endpoints, and the extension to morphologically diverse samples remain open.

This work uses data from the SPARC database [4]. The per-galaxy profiling was developed in collaboration with ChatGPT/5.5 (OpenAI). The theoretical prediction was developed within the SFT framework [8]. The author thanks Claude/Opus (Anthropic) for contributions to the narrative development.

References

- [1] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Phys. Rev. Lett.* **117**, 201101 (2016).
- [2] M. Milgrom, *Astrophys. J.* **270**, 365 (1983).
- [3] J. F. Navarro *et al.*, *Mon. Not. R. Astron. Soc.* **471**, 1841 (2017).
- [4] F. Lelli, S. S. McGaugh, and J. M. Schombert, *Astron. J.* **152**, 157 (2016).
- [5] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020).
- [6] J. M. Schombert, S. S. McGaugh, and F. Lelli, *Mon. Not. R. Astron. Soc.* **483**, 1496 (2019).
- [7] B. Famaey and S. S. McGaugh, *Living Rev. Relativ.* **15**, 10 (2012).
- [8] S. Fosmark, *Solvency Field Theory V11: Master Framework*, Zenodo (2026). <https://zenodo.org/records/20350797>